National police crash files (BAAC), 2019–2025
1
Data
Characteristics
Locations
Vehicles
Users
Linked record
one row per rider
Analysis sample
46,920 riders · 44,817 crashes
2
Variable ordering
Tier 0
Time and place
Tier 1
Rider and vehicle
Tier 2
Road and conditions
Tier 3
Crash configuration
Outcome
killed or hospitalized
Each factor is adjusted only for variables in the same or earlier tiers
3
Validation design
2019
sensitivity
2020–2023 · development · 29,208 riders
cross-validation grouped by crash
2024–2025 · locked test
17,712 riders · evaluated once
4
Estimation
Prediction
gradient boosting · logistic regression
nine validation protocols
Explanation
TreeSHAP attributions
stability over 25 refits
Causal estimation
double machine learning
13 pre-specified factors
5
Audit
Audit of SHAP against double machine learning
agreement in sign · magnitude · rank
6
Stability
Sources of disagreement
adjustment set · predictive model
Heterogeneity
group average effects
Robustness
12 checks · unobserved confounding